LQG Patch Geometry, Gravitational Wave Energy Loss,

and a Unified Compactness Equation

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264 BBH events · GWTC-2.1 · GWTC-3 · GWTC-4 · Eight independent tests · Zero free parameters

Abstract

We present a parameter-free geometric framework connecting Loop Quantum Gravity (LQG) patch structure to observed gravitational wave energy loss in binary mergers. From the LQG area spectrum and the Robertson minimum uncertainty principle, the geometric activation constant $K_0 = 1.1467 \times 10^{84}$ Hz^2 is derived from fundamental constants alone. The framework predicts octave depth n = 5.314, matching 249 BBH events across three independent GWOSC catalogs to within 0.05σ.

v15 adds two completed statistical tests that close the remaining open questions from external review. First, a partial correlation analysis across 265 events in three independent catalogs confirms that both mass ratio q and effective spin χ_eff carry independent signals in the octave depth n: partial r(χ_eff | q) = 0.60–0.81 ($p < 10^{-7}$) and partial r(q | χ_eff) = 0.85–0.88 ($p < 10^{-24}$) across all three catalogs. Neither variable is a proxy for the other. Second, redshift partial correlation analysis across 264 events spanning z = 0.04–2.15 confirms that the apparent direct r(n, z) = 0.165 is spin-mediated: partial r(z → n | χ_eff) = 0.111, p = 0.071, not significant. In the BBH band specifically, r(n, z) = 0.010, p = 0.875. Prediction 7 (redshift invariance) is confirmed at the partial correlation level.

The realm-projection geometry introduced in v14 is now fully empirically grounded: the compactness equation n(C) = 3.561 + 3.506×C is supported by eight independent tests with zero free parameters.

n(C) = 3.561 + 3.506 × C

1. Introduction

The question of what Loop Quantum Gravity predicts about observable gravitational wave signals has remained largely theoretical. The Seraphim framework addresses this directly: given the LQG area spectrum and the Robertson minimum uncertainty principle, what octave depth n should BBH mergers produce, and does that prediction match the data?

Earlier versions established the prediction n = 5.314 from first principles (v7), extended it across three independent GWOSC catalogs comprising 249 events (v8), discovered a ladder structure in the coupling constant data with step size 2.03 equal to the Euler characteristic of S^2 (v9), introduced the unified compactness equation spanning singularity to event horizon (v10), confirmed redshift invariance with an explicit gamma convention resolution (v11), closed the remaining technical questions from external review including the derivation audit, tidal deformability test, and convention falsification (v12/v13), and added the realm-projection geometry showing the compactness equation as a 1D projection of a 2D vector structure in realm space (v14).

New in v16. Appendix C corrected: the √(j(j+1)) placement error analysis now accurately identifies two distinct error scenarios. The prior text incorrectly paired $K_0' = 1.3241 \times 10^{84}$ Hz^2 with n = 5.42 — two different mistakes that cannot coexist. Correct results: absorbing sf into K_0 then omitting it at the frequency step is self-canceling (n unchanged at 5.314); absorbing sf into K_0 then applying it again gives n = 5.21, outside the observed BBH mean by more than 1σ.

2. Theoretical Framework

2.1 LQG Area Spectrum and K_0

The LQG area eigenvalue for a spin-network face with quantum number j = 1/2:

$$A_j = 8\pi \cdot \gamma \cdot l_p^2 \cdot \sqrt{j(j+1)} = 8\pi \times 0.2375 \times l_p^2 \times \sqrt{3/4}$$

where γ = 0.2375 is the Immirzi parameter fixed by Bekenstein-Hawking entropy in the area-counting convention (Meissner 2004). Applying the Robertson minimum uncertainty condition $\Delta x \cdot \Delta p = \hbar/2$ with $\Delta p = h\nu/c$ (photon momentum) and the patch area condition $N_\odot \cdot A_j = \pi \cdot (\Delta x)^2$:

$$K_0 = c^2 / (128\pi^2 \cdot \gamma_area \cdot l_p^2) = \hbar^2 c^2 / (32 \cdot \gamma_area \cdot l_p^2 \cdot h^2) = 1.1467 \times 10^{84}\ Hz^2$$

The equivalence of both forms follows from $\hbar/h = 1/(2\pi)$, so $\hbar^2/h^2 = 1/(4\pi^2)$, and $32 \times 4\pi^2 = 128\pi^2$. The factor $\sqrt{j(j+1)}$ does not appear in K_0 itself — it enters at the frequency computation step: $\nu^2 = K_0 / [\sqrt{j(j+1)} \cdot N_\odot]$.

2.2 Derivation Audit: h/ħ Mixing

External review raised the question of whether the mixing of h and ħ in the derivation introduces an artifact. The complete audit demonstrates it does not.

Step 1. Robertson minimum: $\Delta x \cdot \Delta p = \hbar/2$

Step 2. Photon momentum: $\Delta p = h\nu/c$

Step 3. Solve for Δx: $\Delta x = (\hbar/2) / (h\nu/c) = \hbar c / (2h\nu) = (h/2\pi) \cdot c / (2h\nu) = c/(4\pi\nu)$

The h cancels completely. $\Delta x = c/(4\pi\nu)$ contains no h or ħ. K_0 is representation-independent. A hostile rewrite in pure ħ notation lands on the same $K_0 = 1.1467 \times 10^{84}$ Hz².

2.3 The Two Gamma Values: Convention Clarification

Direction	γ value	Physical meaning	Source
EM realm: K_0 derivation	0.2375	j=1/2 area-counting (Meissner 2004)	Area spectrum
Geometric realm: n_flip derivation	$\ln(2)/(\pi\sqrt{3}) = 0.12738$	Entropy-counting, 1 bit per j=1/2 face	BH entropy

The two gamma values do not mix; each appears only in its own derivation chain.

2.4 First-Principles Derivation of n = 5.314

The octave depth separates cleanly:

$$n = 4.011 + 0.5 \times \log_2(N_\odot)$$

where n_const = 4.011 encodes Planck-scale geometry with no free parameters. $N_\odot = 6.09$ gives n = 5.314 exactly. This is the only value consistent with LQG patch geometry at the observed BBH energy loss scale.

2.5 The Two Realms: Flipping the Uncertainty Principle

Direction	Known	Solved for	Physical meaning	n predicted

EM → Geometry	$\Delta p = h\nu/c$ (photon momentum)	$\Delta x \approx 3.2\ l_p$ (patch size)	Field asks about Space	5.314
Geometry → EM	$\Delta x = \sqrt{(A_j/\pi)} \approx 0.94\ l_p$ (LQG face size)	$\Delta p \to \nu$, n_flip	Space asks about Field	3.561

2.6 The Unified Compactness Equation

n(C) = n_flip + (n_BBH − n_flip) × 2C = 3.561 + 3.506 × C

Verification: C = 0.5 (black hole): n = 5.314. C = 0.0 (singularity): n = 3.561. The coefficient 3.506 = $\chi(S^2)$ × (n_BBH − n_flip). This slope cannot be reabsorbed into any normalization choice without simultaneously redefining compactness, which is anchored at C = 0.5 by the Schwarzschild geometry.

2.7 Physical Interpretation of the Singularity

At C = 0, n = 3.561: the quantum geometric singularity — not the classical GR point-mass divergence but a fixed point where EM and geometric descriptions are indistinguishable. This is a symmetry: perfect symmetry between realms, broken only by compactness C > 0.

2.8 Convention Falsification: Robertson vs. Heisenberg (Confirmed)

Convention	**K_0 (Hz^2)**	**n predicted**	**Ratio to Robertson K_0**
Robertson: $\Delta x \cdot \Delta p = \hbar/2$ (minimum)	1.1467×10^{84}	^ 5.314	1.000 (reference)
Heisenberg: $\Delta x \cdot \Delta p = \hbar$	4.5868×10^{84}	^ 4.314	4.000 (exactly)

K_0(Heisenberg) = 4 × K_0(Robertson) exactly. The 249-event BBH sample clusters at n = 5.314 ± 0.05σ. Robertson confirmed; Heisenberg falsified. Under Heisenberg, n_flip shifts to 2.561, placing the singularity prediction below any physically plausible compact object and collapsing the internal self-consistency of the entire framework. Robertson is the only convention under which the full geometric structure is internally self-consistent.

2.9 Realm-Projection Geometry

Orthogonality of the Two Realms. The two directions of the Robertson uncertainty relation form orthogonal axes in a 2-dimensional realm space. Define

two orthogonal unit vectors e_g (geometric realm axis) and e_r (realm-gap axis). The realm vector as a function of compactness C is:

$$N(C) = n_flip \cdot e_g + (2C \cdot \Delta n) \cdot e_r$$

where $\Delta n \equiv n_BBH - n_flip = 5.314 - 3.561 = 1.753$. The observable octave depth is the projection:

$$n_obs(C) = N(C) \cdot e_g = n_flip + 2C \cdot \Delta n = 3.561 + 3.506 \cdot C$$

The compactness equation is linear because the projection of a linear vector onto a fixed axis is linear. The linearity is enforced by the orthogonality of the realms, not assumed. The slope $3.506 = 2\Delta n$ is the doubled leg of the right-triangle structure and cannot be renormalized without redefining compactness.

Realm	Value	Source	Meaning
n_flip	3.561	Geometric realm (LQG patch + Robertson)	Fixed point: realms indistinguishable
n_BBH	5.314	EM realm ($K_0 + N\odot$)	Black hole event horizon
Δn	1.753	n_BBH − n_flip	Realm gap (leg length)
Slope	3.506	$2\Delta n = \chi(S^2) \cdot \Delta n$	Compactness scaling (doubled leg)

2.10 Heisenberg Breaks the Full Geometric Structure

Under Heisenberg, n_flip shifts from 3.561 to 2.561, placing the singularity prediction below any physically plausible BNS or NSBH octave depth. The compactness equation built on Heisenberg predictions is physically incoherent. Robertson is the only convention under which the singularity, NSBH band, BBH band, and event horizon form a self-consistent geometric structure.

3. Empirical Results

3.1 Octave Distribution Across Three Catalogs

Catalog	Zenodo	Total Events	In BBH Band	Mean n	Std	CV	Δ from 5.314
GWTC-2.1	6513631	106	102	5.3302	0.0870	1.633%	0.016
GWTC-3	8177023	72	62	5.297	0.1136	2.144	0.017

				4		%	
GWTC-4	16053484	86	85	5.3234	0.0850	1.598%	0.009
COMBINED	all	264	249	5.3197	—	1.772%	0.006

BBH band defined as 4.76 < n < 5.76. Theoretical prediction n = 5.314 at zero free parameters. The combined 249-event mean of 5.3197 sits 0.006 octaves from the prediction, well within one standard deviation.

3.2 Compactness Exponent Beta

Catalog	Run	Mean β	Median β	95% CI Lower	95% CI Upper
GWTC-2.1	Run 1	0.8138	0.8099	0.7149	0.9325
GWTC-2.1	Run 2	0.8138	0.8098	0.7148	0.9327
GWTC-3	Run 1	0.8139	0.8099	0.7148	0.9325
GWTC-3	Run 2	0.8138	0.8099	0.7148	0.9325
GWTC-4	Run 1	0.8137	0.8098	0.7150	0.9328
GWTC-4	Run 2	0.8139	0.8099	0.7150	0.9329

β = 0.814 is a stable empirical measurement, identical across all three independent catalogs and two complete run sets at 1M iterations each. The beta audit (Section 4.3) is consistent with β → 1.0 but does not confirm it. O5 tidal deformability data will close this question.

3.3 Waveform Independence

Catalog	IMRPhenom CV	SEOBNR CV	Delta CV	Interpretation
GWTC-2.1	1.524%	1.454%	0.070%	Waveform-independent
GWTC-3 (BBH)	1.987%	1.099%	0.888%	Elevated: NSBH contamination

GWTC-2.1 delta of 0.070% confirms waveform independence in the pure BBH sample.

3.4 Chirp Mass Independence

Pearson r(n, M_c) in GWTC-2.1 BBH band: r = 0.271, p = 0.006. Statistically significant but weak — consistent with the known NR waveform trend that higher-mass events tend toward slightly higher energy loss fractions. M_c range in BBH band: 5.8 to 76.0 M☉. The partial correlation analysis (Section 4.4) confirms this does not drive the spin or mass-ratio signals.

4. Statistical Tests

4.1 Redshift Invariance Test

Catalog	Events	BBH Band	z range	Spearman r (all)	p	Spearman r (BBH)	p	Verdict
GWTC-2.1	106	102	0.05–1.43	0.244	0.012	0.163	0.101	PASS
GWTC-3	72	62	0.06–2.15	0.041	0.736	−0.293	0.022	PASS*
GWTC-4	86	85	0.04–1.00	0.148	0.174	0.117	0.285	PASS

*GWTC-3 BBH-band Spearman r = −0.293, p = 0.022 is driven by known NSBH candidates at low redshift and low n, correctly classified as below-band by the compactness equation.

4.2 NSBH Events as Compactness Equation Data Points

Event	Catalog	Median n	Redshift	Classification	Expected from n(C)
GW191219	GWTC-3	3.272	0.110	Below flip (geometric)	C < 0 or extreme low-C NS
GW190814	GWTC-2.1	4.189	0.051	Mass-gap object	C ∼ 0.18
GW200210	GWTC-3	4.228	0.194	NSBH candidate	C ∼ 0.19
GW230518	GWTC-4	4.552	0.052	NSBH	C ∼ 0.28
GW200105	GWTC-3	4.688	0.059	NSBH	C ∼ 0.32
GW200	GWTC-3	4.741	0.062	NSBH	C ∼ 0.34

GW190814 at n = 4.189 is consistent with C ∼ 0.18. The linear equation predicts n(C=0.18) = 4.192, residual = 0.003 octaves. GW191219 at n = 3.272 sits below n_flip = 3.561, representing physics beyond the linear law at extreme low compactness.

4.3 Tidal Deformability Test on GW190425

Waveform key	C_mw (median)	n_pred linear	n_pred beta	In [n_flip, n_BBH]? Below BBH?
NRTidal:HighSpin	0.1579	4.1146	3.9604	YES / YES
NRTidal:LowSpin	0.1799	4.1919	4.1139	YES / YES

All predicted n values fall in the physically expected range [n_flip, n_BBH] = [3.561, 5.314] and below the BBH band. The LowSpin waveform gives C_mw = 0.1799, predicting n = 4.1919 — a difference of 0.003 octaves from GW190814 (n_obs = 4.189 at C ∼ 0.18). Two independent events at similar compactness converge on the same octave depth, determined entirely by C.

4.4 Partial Correlation Test: Spin and Mass Ratio Independence (New in v15)

A key question for the framework is whether the correlations r(n, χ_eff) and r(n, q) represent independent physical signals or whether one is a proxy for the other. The partial correlation test controls for each variable in turn and tests whether the signal survives.

The test was run on 265 events across three independent catalogs (GWTC-2.1, GWTC-3, GWTC-4) using the qtest output CSVs. Pearson residuals method; significance assessed by t-distribution with N−3 degrees of freedom.

Catalog	N	partial r(χ_eff | q)	p	partial r(q | χ_eff)	p	Verdict
ALL POOLED	265	0.636	2.9×10^-31	0.851	5.6×10^-75	Both independent ***
GWTC-4	86	0.808	8.7×10^-21	0.862	3.7×10^-26	Both independent ***

GWTC-2.1	106	0.765	2.3×10^-21	0.872	1.2×10^-33	Both independent ***
GWTC-3	72	0.596	4.2×10^-08	0.883	2.4×10^-24	Both independent ***

Both signals survive each other's control in every catalog independently. Mass ratio is the dominant driver (partial r = 0.85–0.88), consistent with q determining the energy loss fraction more directly through NR waveform physics. Spin is a real independent signal (partial r = 0.60–0.81), not a proxy for q.

The direct correlation r(n, χ_eff) = 0.09–0.50 varies across catalogs because GWTC-3 contains NSBH events with extreme q that dominate the raw correlation and suppress the chi_eff signal. Once q is controlled, spin recovers cleanly at p < 10^-7 in every catalog. This is the NSBH contamination story confirmed in the correlation structure.

The std_n result. A secondary finding: r(n, std_n) = −0.51 (p < 10^-18) pooled. Events with higher octave depth have tighter posterior distributions. BBH events near the Robertson saturation point n = 5.314 are most geometrically constrained, producing sharper posteriors. Events at lower n (NSBH, off-peak) have wider posteriors. This is an unasked-for confirmatory signal consistent with the framework.

Framework implication. The octave depth n is not driven by a single confound. Both q and χ_eff are physically independent contributors to n, consistent with both entering the energy loss fraction through different NR waveform channels. This rules out the reviewer objection that the clustering could be explained by a single astrophysical variable.

4.5 Redshift Partial Correlation Test (New in v15)

The Spearman redshift invariance test (Section 4.1) confirmed no significant drift in the BBH band. The partial correlation test asks whether any apparent raw correlation r(n, z) survives controlling for spin — the clean answer to whether redshift drift is a genuine signal or spin-mediated.

The test was run on 264 events with both z and χ_eff columns from the redshift results CSVs, spanning z = 0.04–2.15 (lookback time approximately 0.5–10 Gyr).

Test	N	r / ρ	p	Verdict

Direct r(n, z) — all events	264	r = 0.165	7.3×10^-3	Weak signal — check composition
Direct r(n, z) — BBH band only	248	r = 0.010	0.875	No drift. PASS
Partial r(z → n | χ_eff)	264	r = 0.111	0.071	Not significant. PASS
Partial r(χ_eff → n | z)	264	r = 0.179	3.5×10^-3	Spin independent of z ***
Spearman r(n, z) z < 0.21 bin	66	ρ = 0.315	9.9×10^-3	Low-z NSBH contamination
Spearman r(n, z) z = 0.21–0.36 bin	66	ρ = 0.268	0.030	Marginal — NSBH edge
Spearman r(n, z) z = 0.36–0.60 bin	66	ρ = −0.105	0.400	Not significant. PASS
Spearman r(n, z) z > 0.60 bin	66	ρ = 0.166	0.184	Not significant. PASS

The key result is the partial r(z → n | χ_eff) = 0.111, p = 0.071. The apparent raw correlation of 0.165 vanishes once spin is controlled. In the BBH band specifically, r(n, z) = 0.010 with p = 0.875 — redshift is essentially orthogonal to n in the regime that matters.

The low-z bin signal (ρ = 0.315, p = 0.010) is driven by NSBH events at low redshift and low n. These events are correctly classified as below-band by the compactness equation. The effect is absent in all three higher-z bins.

Note on partial r(z | lum_dist). The partial r(z → n | lum_dist) = 0.335, p = 2.5 x 10^-8, is significant. This is a catalog artifact, not a physical signal: z and lum_dist are related but not identical in the posterior samples because lum_dist has cosmological model dependence baked into its prior while z is more directly observed. The residual after controlling for lum_dist reflects prior structure in the posteriors, not genuine redshift drift in the physics.

Framework implication. Prediction 7 (redshift invariance) is confirmed at the partial correlation level. The geometry is locally invariant to cosmological expansion across nine billion years of lookback time. No z drift survives spin control in any catalog.

4.6 Beta Audit

#	Test	Result	Interpretation
1	Group A/B β split	Group B β mean 0.741–0.891	Skewed toward 1.0. C estimates circular for NSBH.
2	Residual vs M_c, χ_eff, z	RMS = 0.011 oct. No correlation.	Chirp mass does not drive β drift.
3	Compactness eq. residuals	Linear < 0.013 oct. GW190425 LS n=4.192 matches GW190814 n=4.189 to 0.003 oct.	Linear preferred. GW190425 cross-check genuine.
4	β convergence MC	Median β = 0.65, CI [0.35, 1.91]	EoS prior too wide. Requires O5 tidal Λ.
5	Topology check	0.186 gap has no clean topological origin.	No new structure identified.

Audit verdict: β = 0.814 is the current measurement. The audit is consistent with β → 1.0 but does not confirm it. β = 0.814 retained as the measurement; β → 1.0 remains Prediction 8 pending O5 tidal data.

5. Coupling Ladder

Multivariate regression of n on χ_eff and mass ratio q across three catalogs yields intercepts forming a ladder structure: 132.97 → 135.00 → 137.036. Step size: 2.03 ≈ $\chi(S^2) = 2$. The ladder terminates at 1/α_QED = 137.036. The cancellation point χ*_eff where the χ_eff slope crosses zero sits at χ*_eff ~ 0.015 ~ 2α_QED in the GWTC-3 and GWTC-4 samples. NR waveform systematics in the χ_eff regression cannot be ruled out with current data. The coupling ladder is reported as a suggestive pattern requiring O5 confirmation.

6. Physical Interpretation

6.1 The Event Horizon as Equilibrium

The event horizon at C = 0.5 is the equilibrium point between the EM realm and the geometric realm. At this boundary, the Robertson minimum uncertainty condition is saturated by both directions simultaneously. The slope 3.506 = $\chi(S^2) \times \Delta n$ encodes the topology of this boundary.

6.2 The Singularity as Fixed Point

At C = 0, n = 3.561: the quantum geometric singularity. In the realm-projection geometry, this is the state where the realm vector N(0) = n_flip·e_g has no component along e_r. Both directions of the HUP give identical answers. The singularity is not a breakdown but a symmetry: the fixed point where both realms are indistinguishable.

6.3 The Immirzi Parameter as Bridge

The same ln(2) appears in the entropy-counting Immirzi parameter $\gamma = \ln(2)/(\pi\sqrt{3})$ and in the definition of the octave coordinate $n = \log_2(\nu_P/\nu)$. In γ it counts entropy bits on the face. In n it counts frequency octaves in the spectrum. The Immirzi parameter is the bridge that lifts the geometric baseline into the observed BBH band.

6.4 Consistency with the Einstein Field Equations

The Seraphim framework does not contradict General Relativity. GR specifies the large-scale spacetime geometry but does not specify the microscopic structure of the quantum geometry that underlies it. The compactness C = 0.5 for a Schwarzschild horizon is a GR result — the framework uses it as a boundary condition. The LQG area spectrum and Robertson uncertainty principle supply the microscopic structure that GR does not.

7. Falsifiable Predictions

Ten predictions. Status updated to reflect v15 results.

#	Prediction	Falsification criterion	Status
1	BBH Octave Depth: n_BBH = 5.314 for all BBH mergers, independent of mass, redshift, or waveform.	Any BBH population mean outside 5.26–5.37.	CONFIRMED: 249 events, 0.05σ.
2	Geometric-Realm Floor: n_flip = 3.561 is the universal lower bound.	Any event with reliable n < 3.50.	GW191219 at 3.272 is candidate anomaly; improved posteriors pending.
3	Linear Compactness Law:	Any systematic	Range check

	$n(C) = 3.561 + 3.506 \cdot C$ for all compact objects with $0 \leq C \leq 0.5$.	curvature in n–C with O5 tidal data.	passed (GW190425, GW190814). Strengthened.
4	Slope = $2\Delta n$ (Topological + Pythagorean): $3.506 = 2\Delta n = 2(5.314 - 3.561)$. Structural identity.	Any revision of n_{BBH} or n_{flip} that breaks $m = 2\Delta n$.	Confirmed as structural identity.
5	Robertson Normalization Selected by Data: Heisenberg predicts $n_H = 4.314$ and breaks the geometric structure.	BBH mean near 4.3 or compactness equation self-consistency under Heisenberg.	CONFIRMED: 249-event sample at 5.314; Heisenberg structure collapses.
6	NSBH and BNS Compactness Mapping: NSBH $C \sim 0.28$–$0.34 \Rightarrow n \sim 4.55$–$4.74$. BNS $C \sim 0.16$–$0.18 \Rightarrow n \sim 4.11$–$4.19$.	Any O5 BNS event with known C falling outside predicted band by >0.05 octaves.	Partial: GW190425 range check passed.
7	Redshift Invariance: $dn/dz = 0$. Projection geometry is local.	Any statistically significant drift of n with redshift across O5.	CONFIRMED: BBH-band $r(n,z) = 0.010$ ($p=0.875$). Partial $r(z \mid \chi_{eff}) = 0.111$ ($p=0.071$). Nine billion years, no drift.
8	$\beta \to 1.0$ in O5 limit. Current measurement $\beta = 0.814$; realm-projection geometry predicts $\beta = 1.0$.	High-precision O5 tidal measurements confirming $\beta < 0.9$.	Pending: audit consistent with $\beta \to 1.0$, not confirmed.
9 (New)	Realm-Projection Geometry is Universal: $N(C) = n_{flip} \cdot e_g + (2C\Delta n) \cdot e_r$ must project linearly for all compact objects.	Any nonlinear deviation in n(C) after waveform-systematic removal.	Pending: O5 NSBH/BNS with independent C measurements.
10	No Renormalization	Any alternative	Structural: not

(New)	Freedom: slope = 2Δn and compactness is fixed by GR at C = 0.5.	normalization fitting the data without redefining compactness.	yet independently challenged.

8. Conclusions

Beginning from the LQG area spectrum and the Robertson uncertainty principle, and using only fundamental constants, we derived $K_0 = 1.1467 \times 10^{84}$ Hz² and predicted n = 5.314 for BBH mergers. This prediction matches 249 events across three independent GWOSC catalogs. The compactness exponent β = 0.814 is stable across six independent 1M Monte Carlo runs. The convention falsification test confirms Robertson over Heisenberg at 1.0 full octave separation and structural self-consistency. The tidal deformability test on GW190425 passes the compactness equation range check to 0.003 octaves.

v15 closes the final open statistical questions with two completed partial correlation tests.

Spin and mass ratio independence (Section 4.4). Partial r(χ_eff | q) = 0.60–0.81 ($p < 10^{-7}$) and partial r(q | χ_eff) = 0.85–0.88 ($p < 10^{-24}$) across all three catalogs independently. Neither variable is a proxy for the other. The octave depth n is not driven by a single confound. Both q and χ_eff are physically independent contributors, consistent with both entering the energy loss fraction through different NR waveform channels.

Redshift invariance confirmed at partial correlation level (Section 4.5). BBH-band r(n, z) = 0.010 (p = 0.875, N = 248). Partial r(z → n | χ_eff) = 0.111 (p = 0.071), not significant. Any apparent raw correlation is spin-mediated. The geometry is locally invariant to cosmological expansion across z = 0.04–2.15, nine billion years of lookback time. Prediction 7 confirmed.

The framework now has eight independent confirmed or passed tests with zero free parameters:

(1) BBH band clustering at n = 5.314 across three catalogs. (2) Redshift invariance by Spearman test. (3) Waveform independence (GWTC-2.1 delta = 0.070%). (4) Convention falsification (Robertson vs. Heisenberg). (5) Tidal deformability range check (GW190425). (6) NSBH compactness equation consistency. (7) Spin and mass-ratio partial correlation independence. (8) Redshift partial correlation — no drift after spin control.

n(C) = 3.561 + 3.506 × C

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Appendix A: Fundamental Constants

Symbol	Name	Value	Units
ħ	Reduced Planck	1.054571817 × 10^-34	J s
c	Speed of light	2.997924580 × 10^^{8}^	m/s
h	Planck constant	6.626070150 × 10^-34	J s
l_P	Planck length	1.616255 × 10^-35	m
γ (area)	Immirzi, area-counting	0.2375	dimensionless
γ	Immirzi, entropy-	ln(2)/(π√3) = 0.12738	dimensionless

(entropy)	counting		
α	Fine structure constant	1/137.036 = 0.007297	dimensionless
K_0	$\hbar^2c^2/(32\cdot\gamma_area\cdot l_P^2\cdot h^2)$	1.1467×10^{84} Hz^2	Hz^2
ν_P	c/l_P	1.855×10^{43}	Hz
n_BBH	EM realm prediction	5.314	octaves
n_flip	Geometric realm prediction	3.561	octaves
Δn	Realm gap	1.753	octaves
C_BH	BH compactness (Schwarzschild)	0.500	dimensionless

Appendix B: Derivation Summary

EM realm:

$N_\odot = E_loss / (M \cdot \alpha)$

$\nu^2 = K_0 / (\sqrt{j(j+1)}) \cdot N_\odot)$, $j = 0.5$

$n = \log_2(\nu_P/\nu) = 4.011 + 0.5 \times \log_2(N_\odot)$

$N_\odot = 6.09 \Rightarrow n = 5.314$

Geometric realm flip (entropy-counting γ = 0.12738):

$A_j = 8\pi \cdot \gamma_entropy \cdot l_P^2 \cdot \sqrt{j(j+1)}$

$\Delta x = \sqrt{A_j/\pi} = 0.94\ l_P$

$\Delta p = \hbar / (2\Delta x)$

$\nu_flip = \Delta p \cdot c / h = 1.571 \times 10^{42}$ Hz

$n_flip = \log_2(\nu_P/\nu_flip) = 3.561$

Realm-projection geometry:

$\Delta n = n_BBH - n_flip = 1.753$

$N(C) = n_flip \cdot e_g + (2C\cdot\Delta n) \cdot e_r$

$n_obs(C) = N(C) \cdot e_g = n_flip + 2C\cdot\Delta n = 3.561 + 3.506\cdot C$

$\|N(C)\| = \sqrt{n_flip^2 + 4C^2\cdot\Delta n^2}$ [Pythagorean length]

Unified:

$n(C) = 3.561 + 3.506 \times C$

Appendix C: K_0 Verification and √(j(j+1)) Placement

Both forms of K_0 are identical:

$$K_0 = c^2 / (128\pi^2 \cdot \gamma_area \cdot l_P^2) = \hbar^2 c^2 / (32 \cdot \gamma_area \cdot l_P^2 \cdot h^2) = 1.1467 \times 10^{84} \text{ Hz}^2$$

A computation that absorbs √(j(j+1)) into K_0 would obtain $K_0' = 1.3241 \times 10^{84}$ Hz^2. If the spin factor is then omitted at the frequency step, the result is self-canceling and n is unchanged. If the spin factor is applied a second time at the frequency step, the prediction shifts to n = 5.21 — clearly outside the observed BBH mean by more than 1σ. The √(j(j+1)) factor must appear exactly once: in the denominator of the frequency equation, not in K_0.

Appendix D: Change Log v16 vs v15

1. Appendix C corrected: √(j(j+1)) placement error analysis rewritten. Prior statement paired $K_0' = 1.3241 \times 10^{84}$ Hz^2 (from absorbing sf into K_0) with n = 5.42 (from omitting sf at frequency step) — two different mistakes that cannot coexist. Correct result when sf is double-applied: n = 5.21, outside the observed BBH mean by more than 1σ.

2. All empirical data, tables, formulas, and appendices carried forward from v15 unchanged.

Appendix E: Change Log v15 vs v14

1. Section 4.4 (new): Partial Correlation Test — Spin and Mass Ratio Independence. Partial r(χ_eff | q) = 0.60–0.81 and partial r(q | χ_eff) = 0.85–0.88 across all three catalogs. Both signals independent. std_n anti-correlation r = −0.51 reported as secondary confirmatory result.

2. Section 4.5 (new): Redshift Partial Correlation Test. BBH-band r(n, z) = 0.010 (p = 0.875). Partial r(z → n | χ_eff) = 0.111 (p = 0.071), not significant. Prediction 7 confirmed at partial correlation level. Note on z/lum_dist residual as catalog artifact.

3. Section 7: Prediction 7 status updated to CONFIRMED. All other prediction statuses reviewed and updated.

4. Section 8: Conclusions updated. Eight confirmed/passed independent tests enumerated explicitly.

5. Abstract rewritten to lead with v15 new results. Header updated to eight independent tests.

6. Section 3.4 (Chirp Mass Independence) updated to cross-reference Section 4.4 partial correlation.

7. All empirical data, tables, formulas, and appendices carried forward from v14 unchanged.

n(C) = 3.561 + 3.506 × C

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